

STATEMENT OF WORK

For

Vapor Pressure Analyzer

22 April 2026

1. **Background:**

The Aerospace Fuels Laboratory at Vandenberg Space Force Base (AFPET/PTPLE) provides laboratory quality surveillance and investigative analytical support of flight critical products directly related to the warfighter and space launch operations on behalf of all Air Force and other DoD agencies, as well as commercial entities. AFPET/PTPLE has a vision to create a flexible, dynamic center of excellence that is responsive to customers and strategically positioned and focused to contingency and warfighter operations. We provide the warfighter and space launch activities specialized testing services and capabilities in petroleum, propellants, cryogenics, chemicals and gases for all aerospace vehicles, systems, and equipment.

2. **Scope:**

AFPET/PTPLE has a requirement for one Vapor Pressure Analyzer to perform specification testing on Automotive & Aviation Gasolines (ASTM D4814, ASTM D910), Ethanol Fuel Blends (ASTM D5798) and Jet Fuel (ASTM D1655). This will need to be a durable & robust unit for routine specification and/or investigative testing. Unit must conform to the requirements set forth in ASTM D5191, *Vapor Pressure of Petroleum Products and Liquid Fuels (Mini Method)*.

Note: Copies of ASTM Methods may be obtained by writing to ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshocken, PA 19428-2959; or by telephone at 610-832-9255; or by e-mail at service@astm.org; or through the ASTM website at www.astm.org

3. **Minimum Technical Requirements:**

The following specifications list the minimum requirements for the Vapor Pressure Analyzer:

- 3.1. Shall be one (1) instrument (with any strictly necessary add-ons) capable of automatically running ASTM D5191, and shall meet the following requirements:
 - 3.1.1. Electrically configured to work with standard 100-240V single phase, 50/60Hz electricity. Equipped with 3 wire electrical cord with 20 amp plug
 - 3.1.2. Benchtop unit, Weight ≤ 50 lbs., WxDxH $\leq 24'' \times 24'' \times 24''$
 - 3.1.3. Ability to be used in analysis of Gasoline, Ethanol Fuels, Organic/Aqueous Solvents, and Kerosene-based fuels.

- 3.1.4. Ability to create vacuum without external vacuum pump; automated, moveable piston shall be used to expand the volume of the sample test chamber. Automated sample introduction into chamber.
- 3.1.5. Sample test chamber designed to contain between 5-50 mL of liquid and vapor; Capable of maintaining a vapor-to-liquid ratio from 0.02/1 to 100/1, depending on method & product.
- 3.1.6. Test temperature range of at least 0 – 120 °C, with a stability of at least 0.05 °C. Heating and cooling must be achieved with no external heater or chiller unit required.
- 3.1.7. Test pressure range of 0 – 1,000 kPa, with pressure resolution of at least 0.02 kPa.
- 3.1.8. Equipped with temperature and pressure sensors to provide real-time measurements of these parameters in sample chamber.
- 3.1.9. Sample measurement time of <10 min once testing conditions are achieved. Sample volume of ≤10 mL required.
- 3.1.10. Equipped with built-in touch screen to show & manipulate instrument conditions, methods & results
- 3.1.11. Memory storage to store and retrieve sample information, data & results. Equipped with USB 2.0+, Ethernet & RS-232 ports
- 3.1.12. Ability to connect unit to printer to print results.
- 3.2. Instrument package shall come with mesh inlet filters (80 µm) (Qty:5) and n-Pentane Calibration Standard.
- 3.3. Instrument package shall include a case to store/ship instrument.
- 3.4. Shall include on-site installation, performance verification, and training of instrument at Vandenberg SFB, CA. At least 4 hours of training (not including installation time) to be given for training & questions.
- 3.5. Technical support shall be available for instrument to help identify, troubleshoot and resolve hardware and software problems. Ability to contact by phone or e-mail during business hours, M-F 9-5pm (nominal).
- 3.6. Shall include electronic copies of all manuals and instructions for instrument operation.

4. Qualifying Country Products:

“All products delivered to satisfy this requirement shall be manufactured in a qualifying country source in accordance with Defense Federal Acquisition Regulation Supplement (DFARS) 225.872. Products from other than qualifying countries will not be accepted, and any costs associated with products from other than qualifying countries, including shipping, etc., shall be at the contractor’s expense.”

5. Delivery, Installation, and Lead Time:

- The Contractor shall deliver and install the product within 12 weeks after contract award to the following address:

SHIP TO DoDAAC: FP2075
Aerospace Fuels Laboratory (FP2075)
AFPET/ PTPLE
1747 Utah Ave Bldg 6670
Vandenberg AFB CA 93437-5220

6. Warranty:

The Contractor shall include at least a one (1) year warranty for parts and services.

7. Invoice and Payment:

“The contractor shall be paid upon submission of a properly prepared invoice and after final inspection and acceptance has been made. A proper invoice shall cite the contractor’s name and address, invoice number, invoice date, order number, contract line-item number (CLIN), description of the supplies or service, unit of issue, unit price and extended price, terms of any prompt payment discount offered, and who to notify in the event the invoice is defective. The invoice shall be submitted to in Wide Area Work Flow (WAWF) in accordance with the terms of the final contract award”

8. Delivery Procedures – Commercial Vehicles:

- a. The company shall properly ship the system to the Aerospace Fuels Laboratory at Vandenberg AFB CA. All shipping costs to include insurance will be included in contractor’s proposal. Great care using standard, approved packaging methods shall be taken.
- b. All vehicles larger than a large pick-up truck are required to be inspected by the Vandenberg Space Force Base Commercial Vehicle Delivery Gate (CVDG) prior to entering the installation. Vehicles to be inspected include, but are not limited to, the following:
 1. Step van/panel truck
 2. Tractor/trailer, box and flatbed containing cargo
 3. Tanker trucks
 4. Box trucks
 5. Tour buses
 6. Garbage/recycled waste trucks
 7. Concrete trucks/mixers, dump trucks
 8. Cranes, recreational vehicles, petroleum tanker
- c. This inspection will be conducted at the truck gate (Lompoc Gate).
 1. If the vehicle has the product inside (concrete and asphalt trucks) and timely delivery is necessary due to product deterioration it does not need to enter the CVDG. To bypass the CVDG, the contractor shall submit a list containing drivers’ names, social security numbers and the state in which the driver's license is held for those drivers who will be entering the base. This shall be accomplished 24 hours prior to requested entry time. If the driver's name is not

on the list, he/she will not be allowed access to the installation through these gates and the base will not assume liability for denied access.

2. All irregular deliveries will be sent for inspection. All visitors, which include contractors and commercial vendors without proper background checks and contractor passes, will not be allowed.